10.3 Rockets and Guns

WORKSHEET

The question

STEP III 2017, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s3-q11

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Before a gun fires, the truck moves at V. Afterwards the projectile's velocity is $V - v$ (that is, speed $v - V$ backwards). What is the projectile's velocity *relative to the truck before firing*?

STAGE 1 All at once

Covers: Part (i)

Ask yourself Everything starts at rest, so the total momentum is 0. With $V = 0$, each projectile leaves with velocity $-v$. Write down momentum to find u, then add up the kinetic energies.

STAGE 2 One at a time

Covers: Part (ii)

Pause and try Just before the rth firing, how many projectiles does the truck carry, so what is the moving mass, and what is its speed? Afterwards, what is the speed of the fired projectile? Write down conservation of momentum and rearrange to get $(*)$.

Ask yourself Add up the increments $u_r - u_{r-1}$ from $(*)$. Compare each term with $\frac{mv}{M}$.

STAGE 3 The energy bill

Covers: Part (iii)

Ask yourself Write $K_r - K_{r-1}$: only the truck with its load and the projectile just fired change speed.

Pause and try Factorise the $u_r^2 - u_{r-1}^2$ term as a difference of two squares and use $(*)$ to replace $(M + (n-r)m)(u_r - u_{r-1})$. Then sum over r: the increments telescope.

What's really going on

What should be held fixed?

A real gun, or a rocket engine, throws its projectile at a fixed speed relative to *itself*, measured just after the projectile leaves. In this question the reference is the truck's speed *before* firing. For a small projectile the difference is tiny, but when all n projectiles are fired at once it matters a great deal: in part (i), a heavy load of projectiles is thrown back at speed v as if the truck did not recoil at all, which needs a lot of energy.

If instead every projectile leaves at speed v relative to the truck *after* firing, the conclusion reverses. All at once gives $u = \frac{nmv}{M+nm}$; one at a time gives $u_n = \sum_{k=1}^{n} \frac{mv}{M+km}$, which is larger, because each term is at least $\frac{mv}{M+nm}$. Now firing one at a time wins: each projectile pushes off from a truck that is already moving, and each later gun has less mass to accelerate. (Check this in the exercises.)

The rocket equation

Now let the projectiles become very small and very numerous, with total mass $\mu = nm$ fixed: this is a rocket burning fuel continuously. The sum for u_n becomes an integral (both conventions give the same limit):

$$u_n = \sum_{k=0}^{n-1} \frac{mv}{M+km} \;\to\; \int_0^{\mu} \frac{v\,\mathrm{d}s}{M+s} = v \ln \frac{M+\mu}{M}.$$

This is *Tsiolkovsky's rocket equation*: a rocket whose exhaust leaves at speed v relative to the rocket gains speed

$$\Delta u = v \ln \frac{m_{\text{initial}}}{m_{\text{final}}}.$$

Figure 10.3.1 compares the three answers. The logarithm is the problem: to reach the orbital speed of about 7.8 km/s with an exhaust speed of about 4.5 km/s (good chemical fuel), the mass ratio must be $e^{7.8/4.5} \approx 5.7$, and once air resistance and gravity are included it is nearer 10 or more. Almost all of a rocket is fuel.

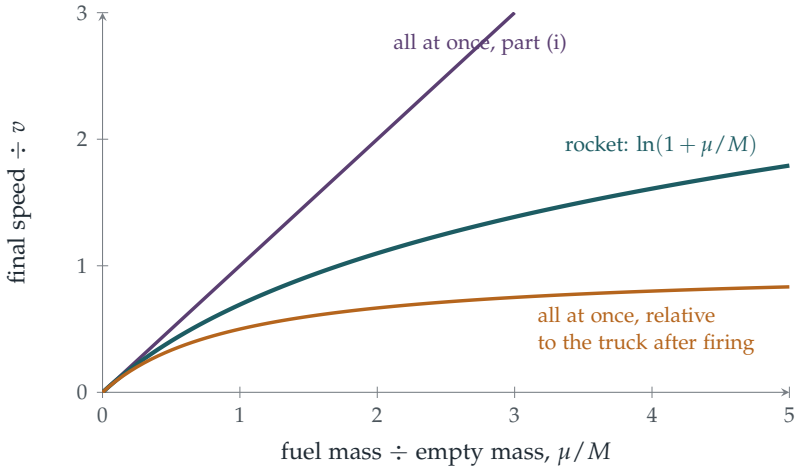


Figure 10.3.1 Final speed against the ratio of fuel to empty mass. Firing everything at once gives μ/M (plum) with the question's convention, but only $\frac{\mu/M}{1+\mu/M}$ (amber) if the exhaust speed is measured relative to the recoiling truck. The rocket, firing continuously, lies in between at $\ln(1+\mu/M)$ (teal). The picture is the inequality $\frac{x}{1+x} \le \ln(1+x) \le x$.

Why rockets have stages

The logarithm also explains staging. A rocket that drops its empty fuel tanks part way up no longer has to accelerate their dead mass. Since the gains of successive stages *add* (like the guns), while each stage's gain depends on the *ratio* of its masses, splitting a rocket into stages lets it reach speeds that a single stage of the same total mass cannot. The Saturn V that took astronauts to the Moon had three stages; its first stage burned about 2000 tonnes of fuel in under three minutes.

Tsiolkovsky

Konstantin Tsiolkovsky, a deaf schoolteacher in the provincial Russian town of Kaluga, published the rocket equation in 1903, in a paper entitled "Exploration of outer space by means of reaction devices". He worked out multi-stage rockets, liquid hydrogen and oxygen fuel, and space stations, decades before any of them could be built. The equation had appeared earlier in less prominent places (William Moore derived a version in 1810), but Tsiolkovsky was the first to see what it meant for spaceflight. "The Earth is the cradle of humanity," he wrote, "but one cannot live in the cradle forever."

Try this next

1 **The other convention.** Redo parts (i) and (ii) supposing each projectile leaves at speed v relative to the truck *after* firing. Show that now firing one at a time gives

the larger final speed.

2 **A comet in a dust cloud.** Try STEP III 2009, Question 11 (step.maths.org/questions/09-s3-q11). It is a variable-mass problem like the rocket, but the comet gains mass instead of losing it. Look out for the speed it tends to.

3 **Two stages.** A rocket has exhaust speed v, empty mass M and fuel mass $9M$. Find its final speed. Now split it into two stages, each with empty mass $\frac{1}{2}M$ and fuel $4.5M$, the first stage being dropped when empty. Compare the final speeds.

4 **The inequality.** Prove that $\frac{x}{1+x} \le \ln(1+x) \le x$ for $x \ge 0$, by comparing $\frac{1}{1+s}$ with 1 and with $\frac{1}{1+x}$ on $[0, x]$.

Further reading

- **Daniel Kleppner & Robert Kolenkow, *An Introduction to Mechanics*** (Cambridge University Press), Chapter 4. Momentum, variable mass and the rocket equation.
- **George Sutton & Oscar Biblarz, *Rocket Propulsion Elements*** (Wiley). The standard engineering text; the opening chapters are readable after this section.
- **Hermann Oberth, *Ways to Spaceflight*** (NASA translation, free online). A pioneer's own account.

The full guided solution is in Section 10.3:
https://beyondstep.keithmcnulty.org/ch10/rockets.html